quantum states where the two regions are unentangled, but there would be an infinite amount of energy at the boundary.

This reasoning extends further. Imagine dividing up all of space into equal-sized boxes. Classically, the field would be doing something in each box, but to avoid infinite energy densities the values must match at the boundaries between boxes. In quantum field theory, therefore, what's happening in one box must be highly entangled with what's happening in neighboring boxes.

That's not all. If a box is entangled with its neighbors, and those neighboring boxes are entangled with their neighbors, it stands to reason that the fields in our original box should be entangled not only with its neighbors, but with the fields one box away. (That's not logically necessary, but it seems reasonable in this case, and a careful calculation affirms that it is true.) There will be a lot less entanglement with the fields one box away than for direct neighbors, but there will still be some there. And indeed this pattern continues all throughout space: the fields in any one box are entangled with the fields in every other box in the universe, although the amount of entanglement becomes less and less as we consider boxes that are farther and farther apart.

That may seem like a stretch, since after all there are an infinite number of boxes in an infinitely big universe. Can the fields in one little region, say, a single cubic centimeter, really be entangled with fields in every other cubic centimeter of the universe?

Yes, they can. In field theory, even a single cubic centimeter (or a box of any other size) contains an infinite number of degrees of freedom. Remember that we defined a *degree of freedom* in Chapter Four as a number needed to specify the state of a system, such as "position" or "spin." In field theory, there are an infinite number of degrees of freedom in any finite region: at every point in space, the value of the field at that point is a separate degree of freedom. And there are an infinite number of points in space, even in just a small region.